IT 239: Web Technology II

BIM 4th Semester

Credits:3
Lecture Hours: 48

Course Description:
This course covers different concepts of server-side Web development using PHP, MySQL, and server-side frameworks.

Course Objectives:
The main objective of this course is to provide students both theoretical and practical knowledge of different technologies that are used for server-side Web development.

Course Details

Unit 1: Introduction **10 LHs**

Server-side Scripting; Setting Up Development Server; PHP Basics: Introduction, Syntax, Comments, Output, Variables, Constants, Data Types, Operators; Control Statements (if, switch, for, for each, while, do-while, break, and continue); Functions (Defining Functions, Passing Arguments, Returning Value from Functions); Including and Requiring Files; PHP Super global.

Unit 2: Strings and Arrays **6 LHs**

String and String Functions; Array (Definition, Numerically Indexed Array, Associative Array, Multidimensional Array, Using for each loop, Multidimensional Arrays, Array Functions); Regular Expressions.

Unit 3: Object-Oriented Programming **10 LHs**

OOP Terminologies; Defining Classes; Creating Objects; Constructor and Destructor; Access Modifiers; Inheritance; Constants; Abstract Classes; Interfaces; Static Methods and Properties; Namespace; Exception Handling.

Unit 4: File and Form Handling **5 LHs**

Reading and Writing Files; Building forms; Retrieving and Processing Form Data.

Unit 5: Working with Database **5 LHs**

Using PHP to Create Connection with Databases, Querying Databases, CRUD Operations Using Forms.

Unit 6: Cookies, Sessions and Authentication **4 LHs**

Setting, Accessing, and Destroying Cookies; HTTP Authentication; Using Sessions (Starting, Ending, Setting a Timeout, and Session Security).

Unit 8: Server-Side Web Framework **8 LHs**

Basics of any one Server-Side Web Framework.

Laboratory Works:
The laboratory work includes creating dynamic web pages using PHP, MySQL, and server-side web frameworks.

Suggested Readings

Learning PHP, MySQL ***and JavaScript: A Step-by-Step Guide to Creating Dynamic Websites***, Robin Nixon, O'reilly.
PHP & MYSQL: ***Server-side Web Development***, Jon Duckett, John Wiley & Sons, Inc.
Murach's PHP and MySQL, Joel Murach and Ray Harris, Mike Murach and Associates, Inc.
Programming PHP: Creating Dynamic Web Pages, Kevin Tatroe , Peter MacIntyre, O′reilly.
www.w3schools.com